HW#4; Problems Statement:

3–13 (a) Formulate mesh-current equations for the circuit in Figure P3–13. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Solve for i_A and i_B.

(c) Use these results to find v_x and i_x.

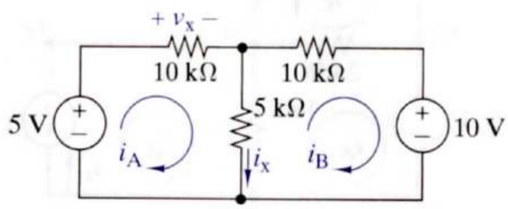

FIGURE P3–13

3–14 (a) Formulate mesh-current equations for the circuit in Figure P3–14. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Solve for i_A, i_B and i_C.

(c) Use these results to find v_x and i_x.

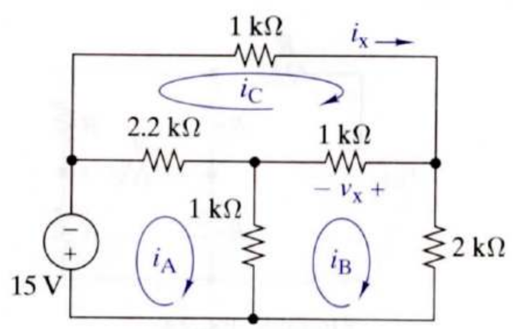

FIGURE P3–14

3–5 (a) Formulate node-voltage equations for the circuit in Figure P3–5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Solve these equations for v_A and v_C.

(c) Use these results to find v_x and i_x.

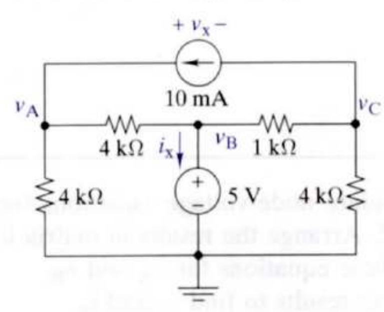

FIGURE P3–5

3–5 **(a)** Formulate node-voltage equations for the circuit in Figure P3–5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_C.
(c) Use these results to find v_x and i_x.

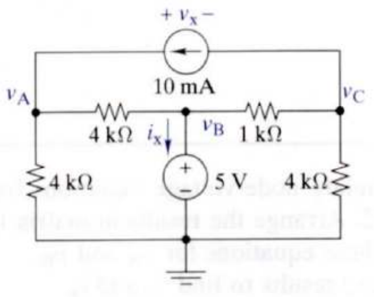

FIGURE P3–5

3–6 **(a)** Choose a ground wisely and formulate node-voltage equations for the circuit in Figure P3–6.
(b) Solve for v_x and i_x when $R_1 = R_2 = R_3 = R_4 = 10\ \text{k}\Omega$, $v_s = 25$ V, and $i_s = 1$ mA.

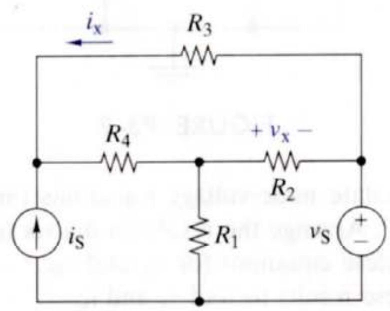

FIGURE P3–6

3–15 **(a)** Formulate mesh-current equations for the circuit in Figure P3–15. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B.
(c) Use these results to find v_x and i_x.

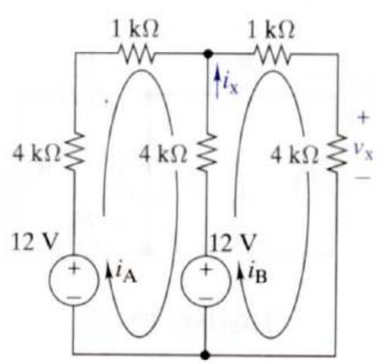

FIGURE P3–15

3–26 **(a)** Formulate mesh-current equations for the circuit in Figure P3–26.

(b) Formulate node-voltage equations for the circuit in Figure P3–26.

(c) Which set of equations would be easier to solve? Why?

(d) Use OrCAD to find the node voltages v_A and v_B and the mesh-currents i_A, i_B and i_C in Figure P3–26.

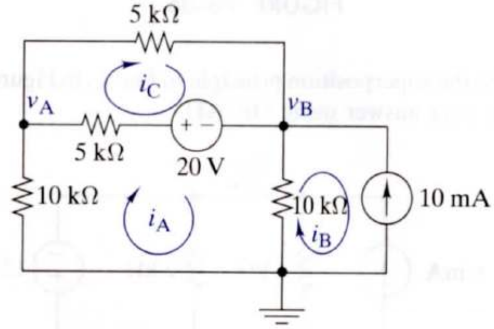

FIGURE P3–26

3–27 **(a)** Formulate mesh-current equations for the circuit in Figure P3–27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(b) Use MATLAB and mesh-current analysis to solve for the mesh currents i_A, i_B, i_C, and i_D in Figure P3–27.

(c) Formulate node-voltage equations for the circuit in Figure P3–27. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

(d) Use MATLAB and node-voltage analysis to solve for the mesh currents i_A, i_B, i_C, and i_D in Figure P3–27 and compare the effort required with each technique, i.e. mesh-current versus node-voltage.

(e) Use OrCAD to verify your results in the previous two parts.

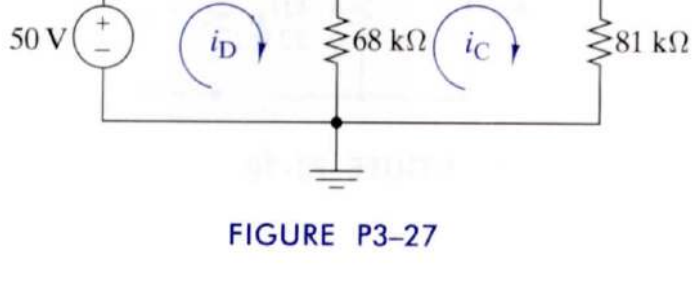

FIGURE P3–27

3–91 E Audio Speaker Resistance-Matchign Network

A company is producing an interface network that they claim would result in an R_{IN} of 600 $\Omega \pm 2$ % and R_{OUT} of 16, 8, or 4 $\Omega \pm 2\%$ – depending on whether the connected speakers are 16, 8, or 4 Ω – selectable via a built-in switch. Their design is shown in Figure P3–91. Prove or disprove their claim.

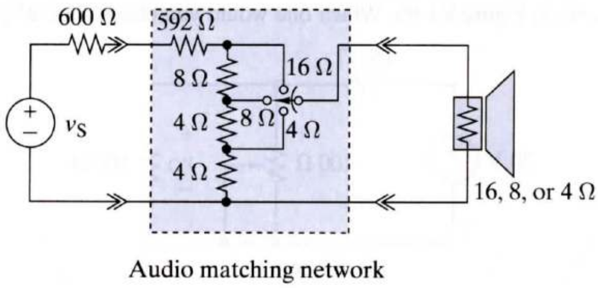

FIGURE P3–91

3–94 D Battery Design

A satellite requires a battery with an open-circuit voltage v_{OC} = 36 V and a Thévenin resistance $R_T \leq 10\ \Omega$. The battery is to be constructed using series and parallel combinations of one of two types of cells. The first type has v_{OC} = 9 V, R_T = 4 Ω, and a weight of 30 grams. The second type has v_{OC} = 4 V, R_T = 2 Ω, and a weight of 18 grams. Design a minimum weight battery that meets the open-circuit voltage and Thévenin resistance requirements.

3–95 E Design Evaluation

A requirement exists for a circuit to deliver 0 to 5 V to a 100-Ω load from a 20-V source rated at 2.5 W. Two proposed circuits are shown in Figure P3–95. Which one would you choose and why?

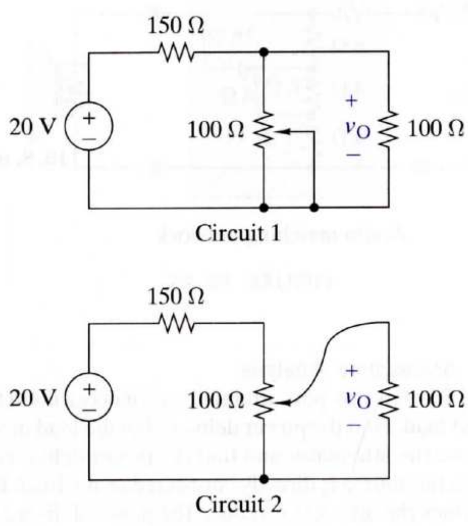

FIGURE P3–95